

UNIT-3

classmate

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TYPES OF SOLIDS:

On the basis of structure, the solid state of matter is classified as:

- ① Crystalline solids (arranged in regular manner)
- ② Amorphous solids (atoms not properly arranged)

CRYSTALLINE

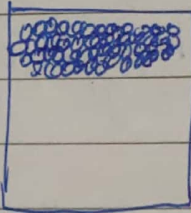
→ arrangement of units of matter is regular & periodic

→ has directional properties & so called anisotropic substances

→ sharp melting point

→ possess regular shape

→ Eg: Cu, Ag, C etc



AMORPHOUS/NON-CRYSTALLINE

→ constituent particles not arranged in an orderly manner. They are randomly distributed

→ not have directional properties & so called isotropic substances.

→ wide range of melting point.

→ do not possess regular shape

→ Eg: glass, plastic, rubber etc.



CRYSTAL LATTICE / SPACE LATTICE:

→ An ideal crystal is composed of atoms or group of atoms arranged in a regular pattern so that atomic arrangement at one side looks exactly same in all respects as the arrangement at some other side.

UNIT CELL:

A smallest group of atoms that repeat itself exactly in all directions in the crystal by means of translation operation 'T' to obtain the entire crystal.

TRANSLATION OPERATION AND TRANSLATIONAL VECTORS:

Translation operation repeats lattice infinite no. of times along a given direction & leaves it unchanged.

It is defined as: $T = n_1 a + n_2 b + n_3 c$ where n_1, n_2, n_3 are arbitrary integers & a, b, c are translational vectors

T operates on 'r' to give 'r'' as follows:

$$r' = r + T \\ = r + n_1 a + n_2 b + n_3 c$$

PRIMITIVE CELL:

a type of unit cell which contains lattice points at corners only & has minimum volume cell.

TYPES OF CRYSTAL:

In 7 crystal systems, there are 14 types of unit cells or lattices (Bravais lattice)

Space lattice can be of following types:

(1) Primitive lattice (P):

lattice pts. are only at corners of unit cell.

(2) Body centered lattice (I):

lattice pts at corners as well as at body centre

(3) Face centered lattice (F):

lattice pts at corners & at the top and bottom base face centres of the unit cell.

(4) Base centered lattice (C):

lattice pts at corners & at the top & bottom base centres of unit cell

7 CRYSTAL SYSTEMS IN 3-D:

- # ① Cubic → simple
→ BCC
→ FCC
- ② Tetragonal → simple
→ BCC
- ③ Orthorhombic → simple
→ BCC
→ FCC
→ Base centred.
- ④ Rhombohedral / Trigonal → simple
- ⑤ Hexagonal → simple
- ⑥ Monoclinic → simple
→ base centred
- ⑦ Triclinic → simple

CRYSTAL PLANES:

Lattice points forming a space lattice have various sets of parallel planes called lattice planes or crystal planes.

MILLER INDICES:

- These are smallest integers used to describe crystal plane.
- defined as reciprocal of the three intercepts on the axes.
reduced to smallest integers (h, k, l)
- no. of indices will match with the dimension of lattice or crystal.

⑦ A plane which is $\parallel$ to any of the coordinate axis has intercept of ∞ on that axis.
 $\therefore$ corresponding miller index is 0.

⑧ All equally spaced $\parallel$ planes have same miller indices i.e. M.I not only defines a particular plane but a set of $\parallel$ planes.

CRYSTAL DIRECTIONS:

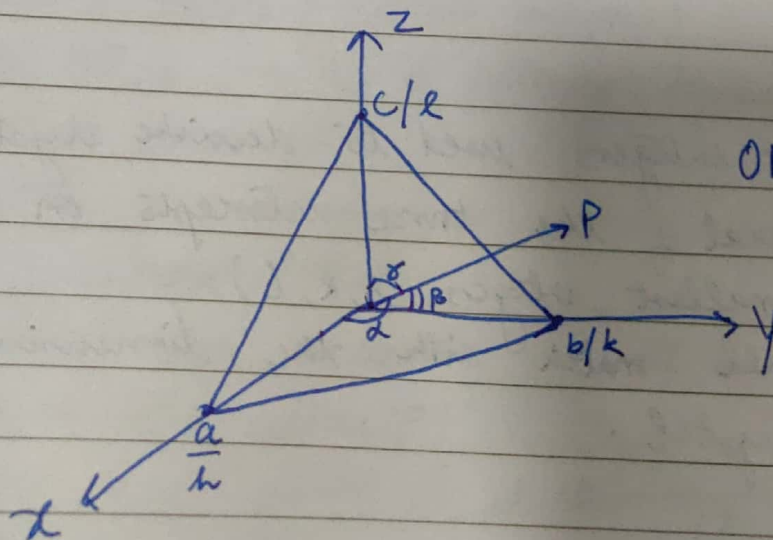
→ to indicate certain directions inside the crystal
 $\rightarrow$ M.I of direction

① Find M.I of plane $\perp$ to that direction

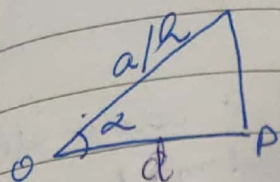
② The $\perp$ plane & direction have the same M.I.

INTERPLANAR SPACING (d):

Consider a family of parallel planes (hkl). The spacing ' d ' b/w the adjacent planes of this family can be calculated by taking any lattice pt as origin & finding the perpendicular distances b/w this origin & the plane which is nearest to this origin. This plane ~~would~~ would have $\frac{a}{h}, \frac{b}{k}, \frac{c}{l}$ as intercepts on the axes resp.



OP is $\perp$ to the plane



$$d = OP = \frac{a}{h} \cos \alpha$$

$$= \frac{b}{k} \cos \beta$$

$$= \frac{c}{l} \cos \gamma$$

Using the law of direction cosines;

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$d^2 \left[\frac{1}{(a/h)^2} + \frac{1}{(b/k)^2} + \frac{1}{(c/l)^2} \right] = 1$$

$$d^2 \left[\left(\frac{h}{a} \right)^2 + \left(\frac{k}{b} \right)^2 + \left(\frac{l}{c} \right)^2 \right] = 1$$

$$d =$$

$$\frac{1}{\sqrt{\left(\frac{h}{a} \right)^2 + \left(\frac{k}{b} \right)^2 + \left(\frac{l}{c} \right)^2}}$$

If $a = b = c$ for cubic lattice,

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

PARAMETERS USED TO DESCRIBE CRYSTAL STRUCTURE:

- ① No. of atoms per unit cell.
- ② Co-ordination no (no. of nearest neighbouring atoms, molecules or ions to a particular atom).
- ③ Atomic radius
- ④ Atomic Packing Fraction (APF): ratio of volume occupied by the atoms or ions in a unit cell to the total volume of the unit cell.

SIMPLE CUBIC STRUCTURE (SCC)

A simple cubic unit cell consists of 8 corner atoms. The contribution of each & every corner atom to unit cell is $1/8$.

$$\therefore \text{Total no. of atoms present in unit cell} = \frac{1}{8} \times 8 = 1$$

$$C.No = 6$$

$$\text{Volume of atoms in unit cell} = \left(\text{no. of atoms in unit cell} \right) \times (\text{volume of 1 atom})$$

$$= 1 \times \frac{4}{3} \pi r^3$$

$$\text{Volume of unit cell} = a^3$$

$$APF = \frac{\frac{4}{3} \pi r^3}{a^3} = \frac{\frac{4}{3} \pi r^3}{(2r)^3} = \frac{4 \pi}{3 \times 8} = \frac{\pi}{6} = 0.52$$

($a = 2r$)

$$\text{Packing fraction} = 52\%$$

BODY CENTERED CUBIC STRUCTURE (BCC):

→ 8 corner atoms + 1 body centered atom

$$\rightarrow \text{Total no. of atoms} = \frac{1 \times 8 + 1}{1} = 2$$

$$\rightarrow C.No = 8$$

$$APF = \frac{2 \times \frac{4}{3} \pi r^3}{a^3} = \frac{2 \times \frac{4}{3} \pi r^3}{\left(\frac{4r\sqrt{3}}{3}\right)^3} = \frac{\left(\frac{8}{3}\pi\right) \left(\frac{\sqrt{3}}{64}\right)}{\frac{\sqrt{3}}{8}} = \frac{\sqrt{3}\pi}{8} = 0.68$$

($a = \frac{4r\sqrt{3}}{3}$)

$$P.F = 68\%$$

FACE-CENTERED CUBIC STRUCTURE: (FCC)

→ 8 corner atoms + 6 face centered atoms

$$\text{total no. of atoms} = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 1 + 3 = 4$$

→ C. No = 12

(4 face centred in each place touched by a corner atom)

($a = 2\sqrt{2}r$)

$$\text{APF} = \frac{4 \times \frac{4}{3} \pi r^3}{a^3} = \frac{4 \times \frac{4}{3} \pi r^3}{(2\sqrt{2})^3} = \left(\frac{16\pi}{3}\right) \left(\frac{1}{16\sqrt{2}}\right) = \frac{\pi}{3\sqrt{2}} = 0.74$$

Packing fraction = 74%

Imp

RELATION B/W DENSITY AND LATTICE CONSTANT (dis b/w 2 lattice pts)

Consider a cubic lattice of lattice constant 'a'. If

$\rho \rightarrow$ density of crystal

$a^3 \rightarrow$ volume of unit cell

then mass of each unit cell = ρa^3 — (1)

If M is the molecular weight & N_A is avogadro's no; then
 mass of each molecule = $\frac{M}{N_A}$

If n is the no. of molecules per unit cell, then
 mass in each unit cell becomes = $\frac{nM}{N_A}$ — (2)

From (1) & (2)

$$\frac{nM}{N_A} = \rho a^3$$

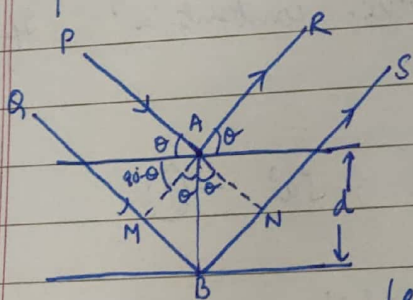
$$a = \left(\frac{nM}{\rho N_A} \right)^{1/3}$$

BRAGG'S LAW: SUMMARY :

	SCC	BCC	FCC
① Volume of unit cell	a^3	a^3	a^3
② Lattice pt. per unit cell	1	2	4
③ Atomic radius (r)	$\frac{a}{2}$	$\frac{\sqrt{3}a}{4}$	$\frac{a}{2\sqrt{2}}$
④ Nearest neighbour distance	a	$\frac{\sqrt{3}a}{2}$	$\frac{a}{\sqrt{2}}$
⑤ C.No	6	8	12
⑥ P.F	52%	68%	74%

BRAGG'S LAW:

Let us consider a set of 2 parallel planes having spacing 'd'. X-rays (of wavelength λ) PA and QB are incident on plane 1 & 2 resp & getting reflected at AR and BS.



Let θ be the angle b/w the incident ray (reflected ray) and the plane.

Let us construct AM and BN $\perp$ on QB and PA

such that $\angle MAB = \angle NAB = \theta$.

As the rays are reflected from planes they will interfere. For constructive interference,

$$\text{path diff b/w two rays} = n\lambda$$

$$\text{Path diff b/w PAR + QBS} = BM + BN$$

$$\text{but } BM = BN = d \sin \theta$$

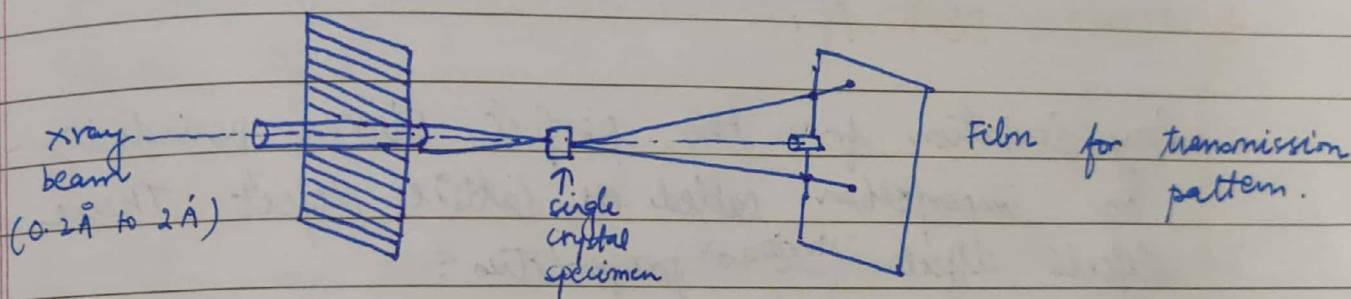
$$\boxed{2d \sin \theta = n\lambda} \text{ which is Bragg's law.}$$

where d = interplanar spacing
 θ = Bragg's angle
 λ = Wavelength of light used.

(*) Bragg planes:

→ The atoms in a crystal may be visualised as families of parallel planes where each family has a certain separation b/w its component planes. These planes are known as Bragg's plane.

LAUE METHOD: (to determine crystal structure)



- ① A single crystal is mounted on a rotating stage & is rotated through certain known angles.
- ② The crystal is irradiated with a polychromatic beam of X-rays.
- ③ The crystal diffracts only those rays & only those wavelengths for which the the planes of interplanar spacing ' d ' & incident angle ' θ ' satisfy Bragg's law i.e. $\boxed{2d \sin \theta = n\lambda}$

A flat-photographic plate is used to receive the reflected / transmitted diffracted beams resulting in laue pattern consisting of a series of spots.

⑩ If sharp - well defined spots are obtained, then the crystal has perfect structure.

* If diffused or broken spots are obtained, then there would be lattice distortion or defects exist in crystal.

⑩ Disadvantage:

or of wide range of wavelengths, each of the wavelength cause different orders of reflection that may superimpose on a single spot.

POINT DEFECTS:

Any deviation from the perfect atomic periodicity is an imperfection called as lattice defect. These defects affect various properties:

- ① Admissibility, diffusion & crystal growth
- ② electrical resistance
- ③ the strength of crystals etc.

→ A point imperfection is localised near a point or atom in the structure. Point defects occur in metallic as well as ionic lattices.

→ does not change the stoichiometry of the crystal.

→ 3 types:

- └ Vacancies
- └ interstitial
- └ isolated impurity.

SCHOTTKY DEFECT:

When the irregularities of the atomic array occurs due to missing of atoms from lattice sites, then a vacancy exists known as schottky defect.

FRENKEL DEFECT:

When an interstitial is caused by transferring an atom from a lattice site to an interstitial site, a vacancy is created. The associated vacancy & interstitial atom is called Frenkel defect.

DIFFERENCES:

- (*) In S.D, ion ends up on the surface of the crystal.
In F.D, the ion is removed to an interstitial position.
- (*) In S.D, the volume of the crystal is increased without any change in mass.
 $\therefore$ Density of crystals decreases.
- F.D, does not change the volume of the crystal, so density remains same.
- (*) Schottky $\rightarrow$ ion missing
Frenkel $\rightarrow$ ion shifts to interstitial position.

DEPENDENCE OF SCHOTTKY DEFECT ON TEMP.

E_v → energy reqd to take an atom from lattice site inside crystal to lattice site on the surface
 nE_v → increase in internal energy associated with production of 'n' isolated atoms vacant sites.

Total no. of ways of choosing n-atoms from crystal containing N atoms is
$$W = \frac{N!}{(N-n)! n!}$$

Increase in entropy = $S = k_B \ln W$

$$= k_B \ln \frac{N!}{(N-n)! n!}$$

This, produces a change in free energy F
 $F = U - TS$

$$= nE_v - k_B T \ln \frac{N!}{(N-n)! n!}$$

$$= nE_v - k_B T \left[N \ln N - N - (N-n) \ln (N-n) + n \ln n \right]$$

$$= nE_v - k_B T [N \ln N - (N-n) \ln (N-n) - n \ln n]$$

Free energy in thermal eq at const. V must be minimum,

$$\left(\frac{\partial F}{\partial n} \right)_T = 0 = E_v - k_B T \ln \frac{N-n}{n}$$

$$0 = E_v + k_B T \ln \frac{n}{N-n}$$

$\ln x!$
 $\approx x \ln x - x$

$$\exp\left(\frac{-E_v}{K_B T}\right) (N-n) = n$$

if $n \ll N$,

$$n \approx N \exp\left(\frac{-E_v}{K_B T}\right)$$

The equilibrium conc. of vacancies $\downarrow$ as temp $\downarrow$

★ Ionic crystals:

Ways of forming n separate pairs $= W = \left[\frac{N!}{(N-n)! n!} \right]^2$

Increase in entropy will be $= S = k_B \ln W$
 $= k_B \ln \left(\frac{N!}{(N-n)! n!} \right)^2$

Corresponding change in free energy

$$F = U - TS$$

$$F = n E_p - K_B T \ln \left(\frac{N!}{(N-n)! n!} \right)^2$$

where E_p = energy of formation of pair

$$F = n E_p - 2 K_B T [N \ln N - (N-n) \ln (N-n) - n \ln n]$$

differentiating above eqⁿ wrt n ,

$$\left(\frac{\partial F}{\partial n} \right)_T = E_p - 2k_B T [0 + \ln(N-n) + 1 - \ln n - 1]$$

$$= E_p - 2k_B T \ln \left(\frac{N-n}{n} \right)$$

Free energy in thermal eq. attained at temp T is constant, so that,

$$E_p - 2k_B T \ln \frac{N-n}{n} = 0$$

$$\ln \frac{N-n}{n} = \frac{E_p}{2k_B T}$$

$$\frac{N-n}{n} = \exp \frac{E_p}{2k_B T}$$

$$n \ll N$$

$$n = N \exp \left(\frac{-E_p}{2k_B T} \right)$$

FRANKEL

— no. of defects in equilibrium at temp T .

Let in a perfect crystal,

E_i = the energy reqd. to displace an atom from regular lattice site to interstitial position.

N_i = the interstitial atoms.

N = the atoms.

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Total ways of forming n defects = $W = \frac{N!}{(N-n)!n!} \cdot \frac{N_i!}{(N_i-n)!n!}$

Increase in entropy,

$$S = k_B \ln W$$

$$= k_B \ln \left[\frac{N!}{(N-n)!n!} \cdot \frac{N_i!}{(N_i-n)!n!} \right]$$

$$= k_B \left(\ln \frac{N!}{(N-n)!n!} + \ln \frac{N_i!}{(N_i-n)!n!} \right)$$

$$= k_B (N \ln N + N_i \ln N_i - (N-n) \ln (N-n) - (N_i-n) \ln (N_i-n) - 2n \ln n)$$

Change in free energy,

$$F = U - TS$$

$$= nE_i - k_B T \ln W$$

differentiate wrt n

$$\left(\frac{\partial F}{\partial n} \right)_T = E_i - k_B T \ln \frac{(N-n)(N_i-n)}{n^2}$$

At eq, free energy is constant;

$$E_i = k_B T \ln \frac{(N-n)(N_i-n)}{n^2}$$

$$\approx k_B T \ln \frac{NN_i}{n^2}$$

$$E_i = k_B T (\ln(NN_i) - 2 \ln(n))$$

$$2 \ln n - \ln NN_i = \frac{E_i}{k_B T}$$

~~$$\ln \frac{NN_i n^2}{NN_i} = \frac{E_i}{k_B T}$$~~

$$n^2 = NN_i \exp\left(\frac{E_i}{k_B T}\right)$$

$$\ln \frac{NN_i}{n^2} = \frac{E_i}{k_B T}$$

$$n^2 = NN_i \exp\left(\frac{-E_i}{k_B T}\right)$$

$$n = (NN_i)^{1/2} \exp\left(\frac{-E_i}{2k_B T}\right)$$